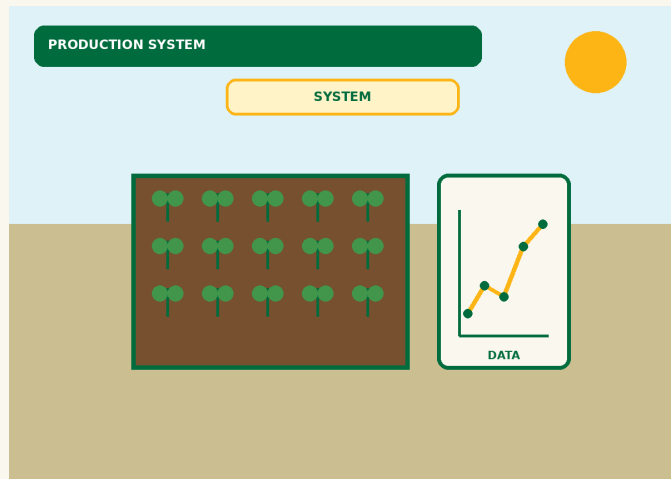




SYSTEM

PRODUCTION SYSTEM

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 601 • PRODUCTION SYSTEM

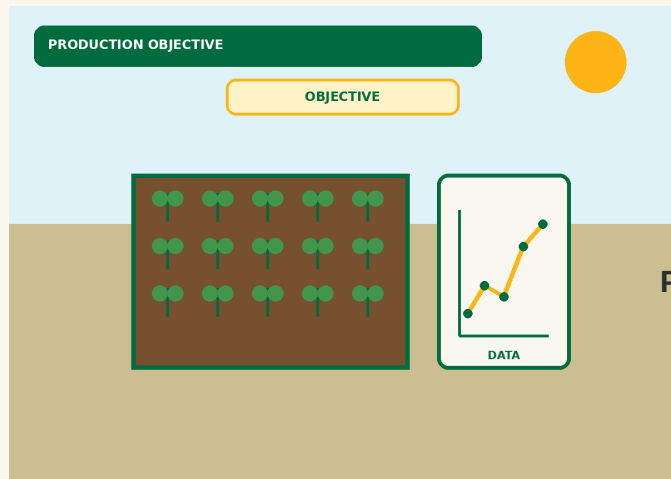
CONCEPT

A crop production system combines land or growing space, inputs, management, labor, and biological processes to produce useful outputs.

INVESTIGATION TASK

Map the main inputs, processes, and outputs of one garden bed.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



OBJECTIVE

PRODUCTION OBJECTIVE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 602 • PRODUCTION OBJECTIVE

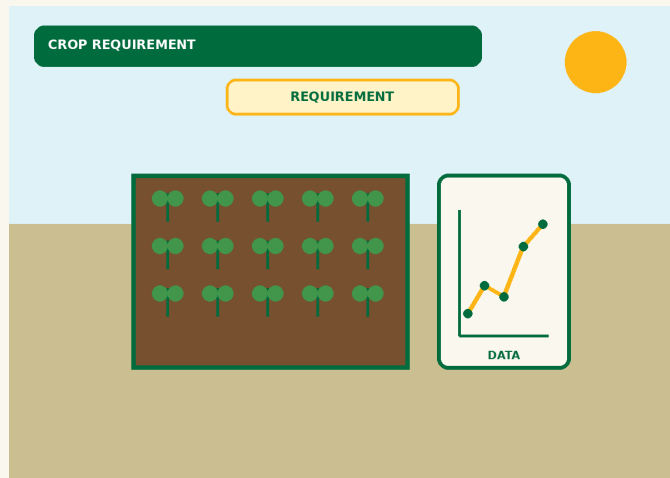
CONCEPT

A production objective states the measurable result a grower intends to achieve, such as yield, quality, timing, or resource efficiency.

INVESTIGATION TASK

Write one measurable objective for the garden.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



REQUIREMENT

CROP REQUIREMENT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 603 • CROP REQUIREMENT

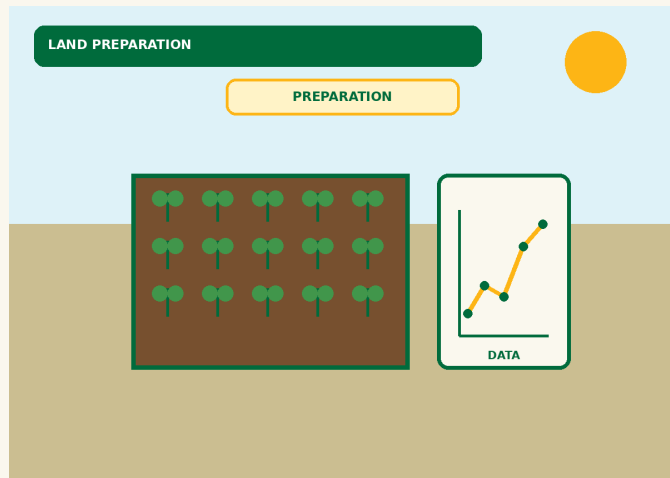
CONCEPT

Crop requirements include suitable water, nutrients, rooting conditions, light, temperature, spacing, and management.

INVESTIGATION TASK

Identify three requirements for one crop.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



PREPARATION

LAND PREPARATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 604 • LAND PREPARATION

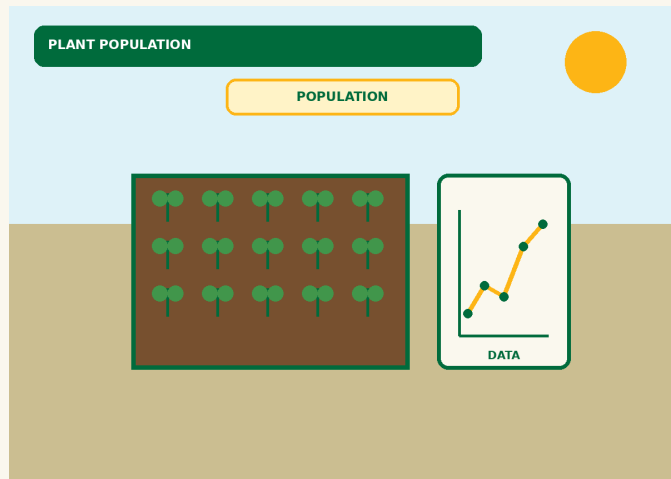
CONCEPT

Land preparation creates suitable physical conditions for establishment, root growth, irrigation, drainage, and field operations.

INVESTIGATION TASK

Evaluate one prepared bed using agreed criteria.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



POPULATION

PLANT POPULATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 605 • PLANT POPULATION

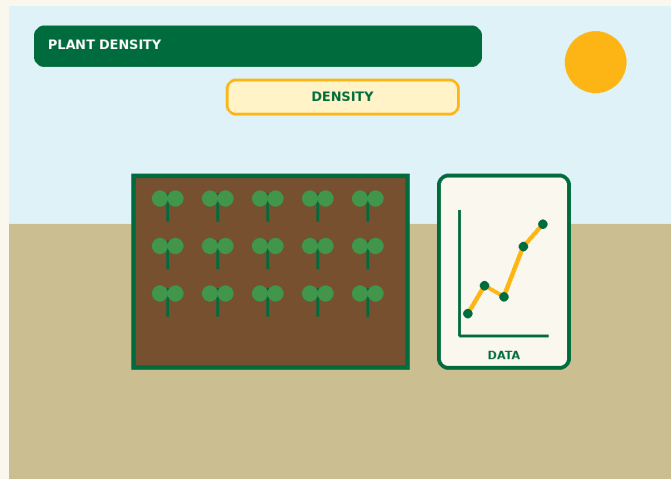
CONCEPT

Plant population is the number of crop plants established within a defined production area.

INVESTIGATION TASK

Calculate plant population in a measured plot.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



DENSITY

PLANT DENSITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 606 • PLANT DENSITY

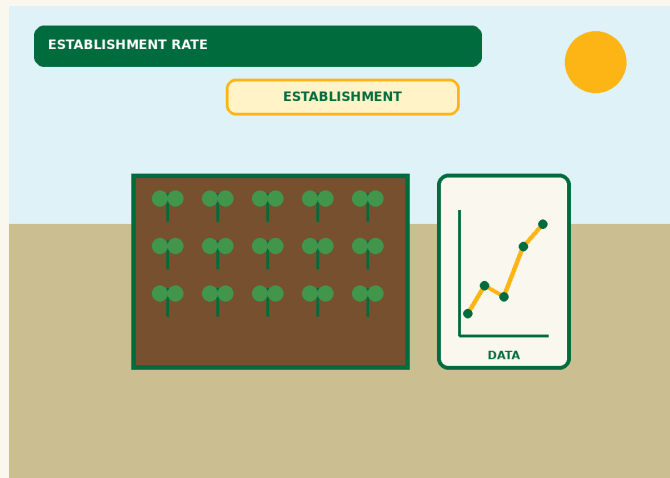
CONCEPT

Plant density expresses the number of plants per unit area and is influenced by row and within-row spacing.

INVESTIGATION TASK

Calculate plants per square metre from field measurements.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



ESTABLISHMENT

ESTABLISHMENT RATE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 607 • ESTABLISHMENT RATE

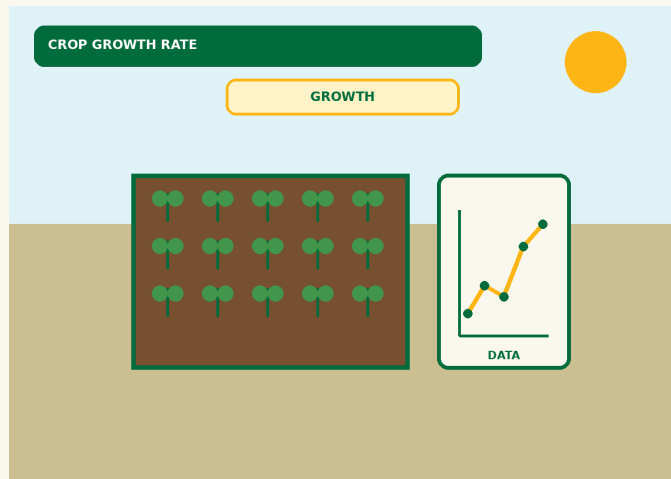
CONCEPT

Establishment rate compares the number of successfully established plants with the number planted or sown.

INVESTIGATION TASK

Calculate establishment percentage for one row.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



GROWTH

CROP GROWTH RATE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 608 • CROP GROWTH RATE

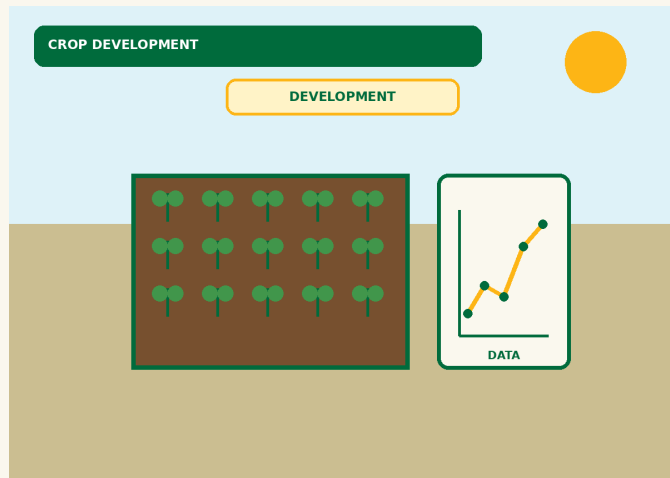
CONCEPT

Repeated measurements can be used to describe how rapidly crop size or biomass indicators change over time.

INVESTIGATION TASK

Calculate change in plant height over a defined period.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



DEVELOPMENT

CROP DEVELOPMENT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 609 • CROP DEVELOPMENT

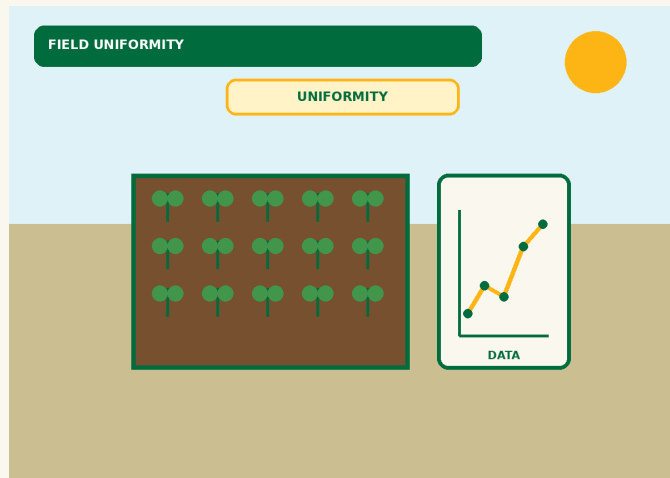
CONCEPT

Development describes progression through stages such as emergence, vegetative growth, flowering, fruiting, and maturity.

INVESTIGATION TASK

Identify the dominant growth stage in a plot.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



UNIFORMITY

FIELD UNIFORMITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 610 • FIELD UNIFORMITY

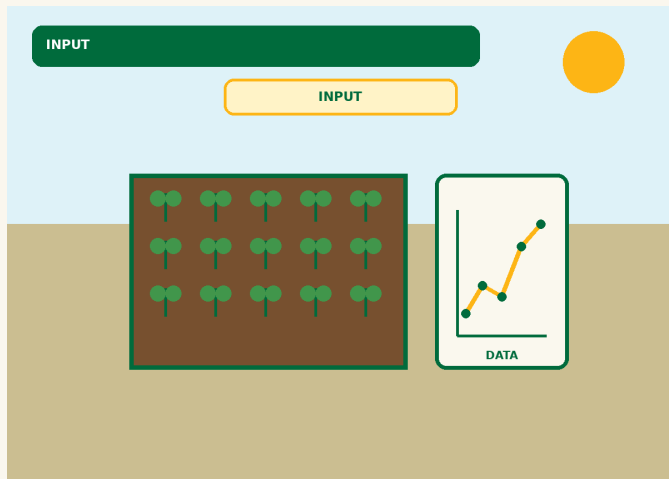
CONCEPT

Field uniformity describes how similar crop establishment, growth, and condition are across a production area.

INVESTIGATION TASK

Compare measurements from several positions.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



INPUT

INPUT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 611 • INPUT

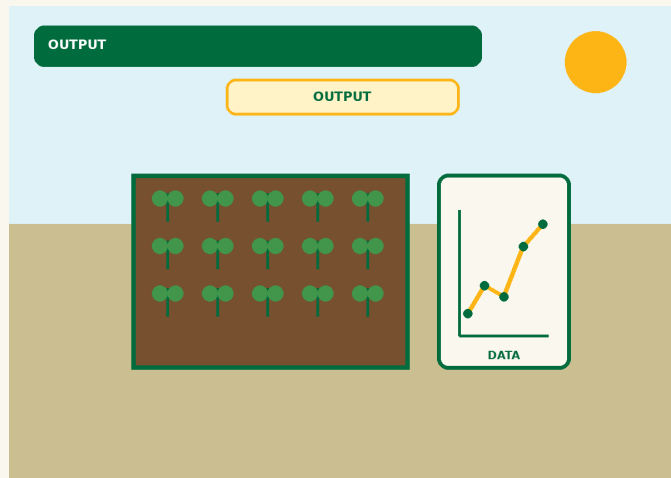
CONCEPT

Inputs include seed, water, nutrients, growing materials, energy, labor, tools, and information used in production.

INVESTIGATION TASK

Classify the major inputs used this week.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



OUTPUT

OUTPUT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 612 • OUTPUT

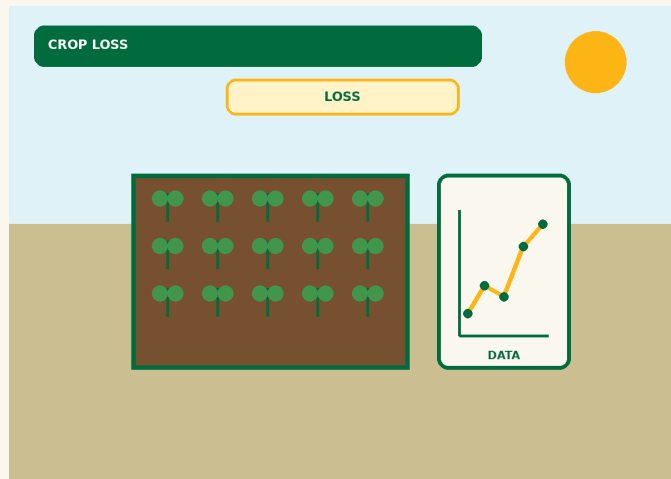
CONCEPT

Outputs include harvested produce as well as residues, data, and other products of the system.

INVESTIGATION TASK

Quantify one useful output from the garden.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



LOSS

CROP LOSS

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 613 • CROP LOSS

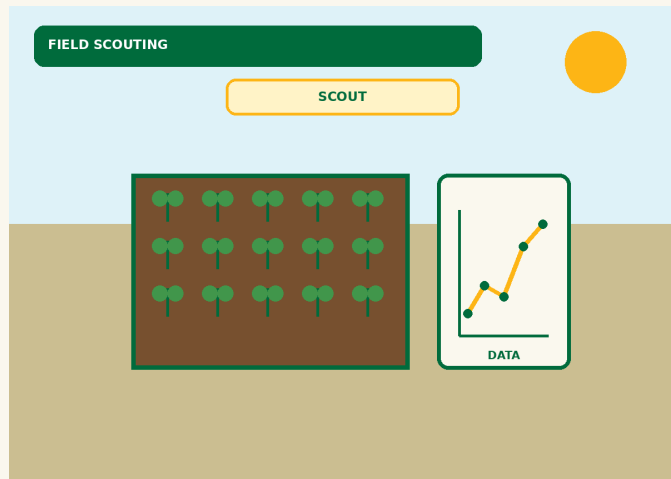
CONCEPT

Crop loss is the reduction in expected production caused by failed establishment, damage, pests, disease, competition, or other constraints.

INVESTIGATION TASK

Estimate missing or damaged plants in a sample area.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



SCOUT

FIELD SCOUTING

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 614 • FIELD SCOUTING

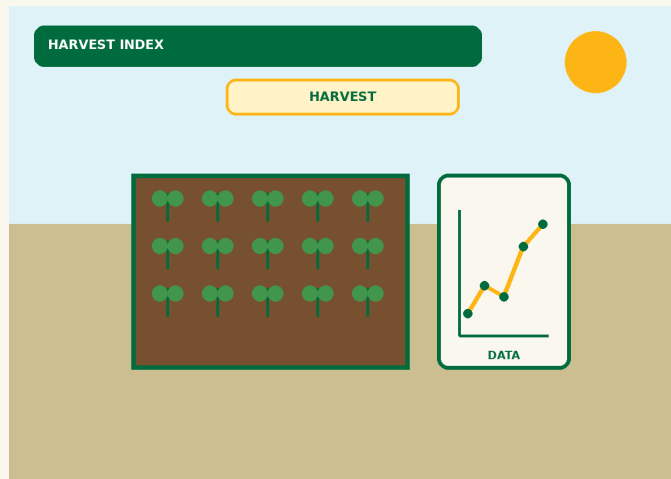
CONCEPT

Field scouting is systematic observation of crop condition and production constraints across a field or garden.

INVESTIGATION TASK

Complete a short scouting route and record findings.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



HARVEST

HARVEST INDEX

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 615 • HARVEST INDEX

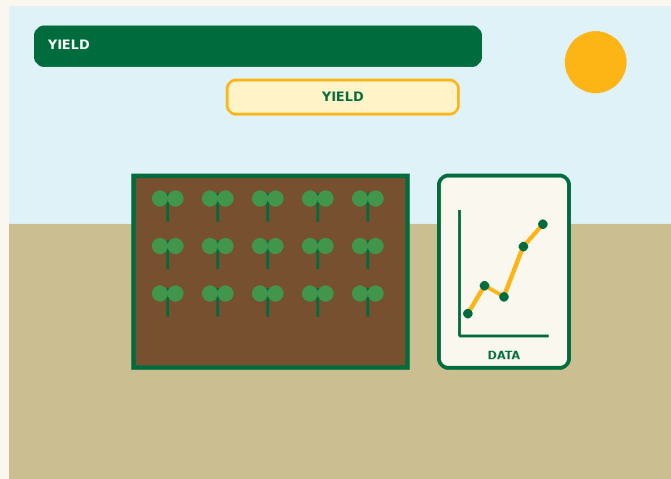
CONCEPT

Harvest records quantify the useful crop material collected from a defined area or population.

INVESTIGATION TASK

Measure harvest from a marked plot.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



YIELD

YIELD

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 616 • YIELD

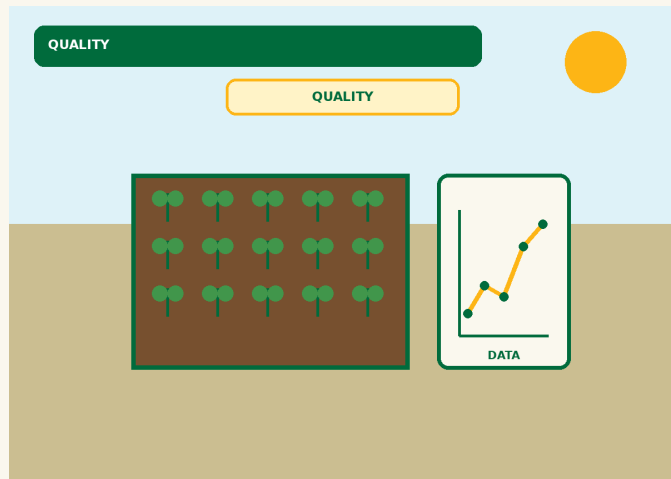
CONCEPT

Yield is the quantity of harvested product per unit area, plant, or production cycle.

INVESTIGATION TASK

Calculate yield per square metre from sample data.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



QUALITY

QUALITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 617 • QUALITY

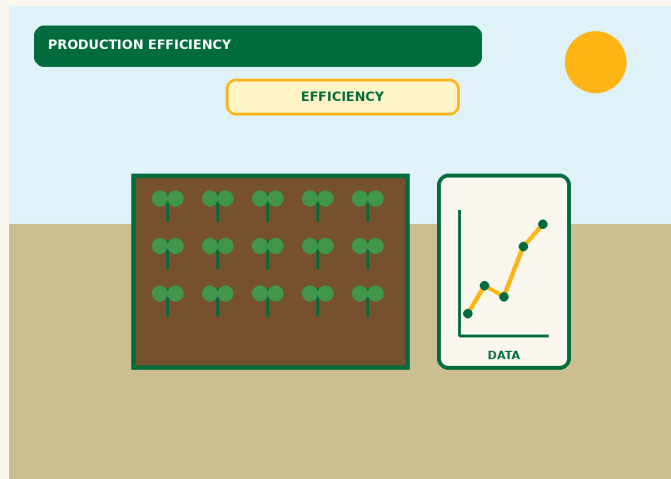
CONCEPT

Crop quality can be evaluated using defined attributes such as size, appearance, maturity, damage, or market specifications.

INVESTIGATION TASK

Score a sample using a simple quality rubric.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



EFFICIENCY

PRODUCTION EFFICIENCY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 618 • PRODUCTION EFFICIENCY

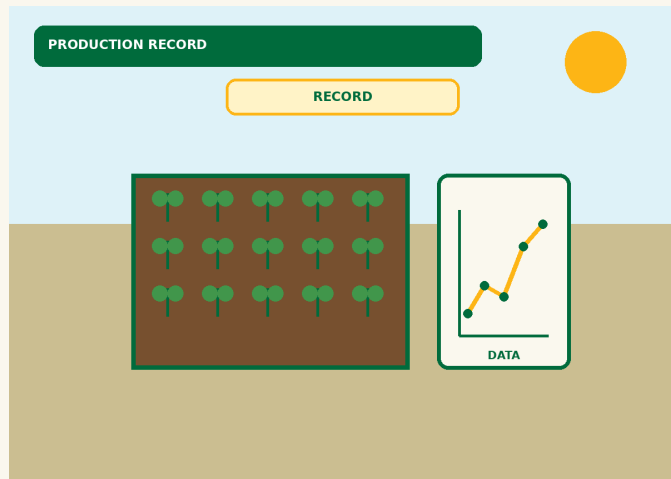
CONCEPT

Production efficiency compares useful output with one or more inputs used to obtain it.

INVESTIGATION TASK

Calculate a simple output-to-input ratio.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RECORD

PRODUCTION RECORD

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 619 • PRODUCTION RECORD

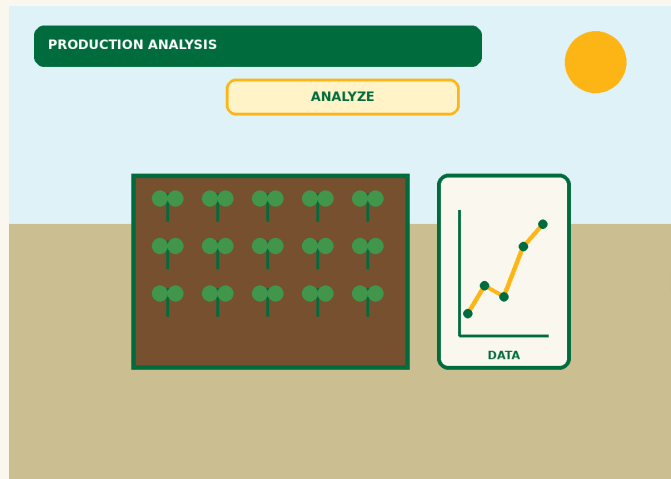
CONCEPT

Production records document dates, inputs, operations, observations, measurements, harvests, and losses.

INVESTIGATION TASK

Audit one record for missing information.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



ANALYZE

PRODUCTION ANALYSIS

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 620 • PRODUCTION ANALYSIS

CONCEPT

Production analysis integrates crop performance, inputs, losses, yield, quality, and records to support management decisions.

INVESTIGATION TASK

Identify one constraint and recommend a measurable response.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE